Architecture.md

Bloom Architecture

Project: Bloom

Document: System Architecture

Version: 1.0

Status: High-Level Engineering Design

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1. Purpose

This document defines the overall system architecture of Bloom.

The objective is to describe how the different components collaborate to transform a learner's career goal into a personalized learning environment.

This document intentionally remains **technology-agnostic**. It focuses on system responsibilities and interactions rather than implementation choices.

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2. Architectural Principles

Bloom is designed around the following principles:

- Modular by design
- AI-first, but not AI-dependent
- Platform-independent
- Scalable
- Explainable
- Privacy-conscious
- Extensible
- Event-driven where appropriate

Every component should have a single clear responsibility.

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3. High-Level Architecture

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                  ┌───────────────────────┐
                  │     User Interface    │
                  │ Browser Extension / Web │
                  └───────────┬───────────┘
                              │
                              ▼
                 ┌─────────────────────────┐
                 │      Backend Gateway     │
                 └────────────┬────────────┘
                              │
       ┌──────────────┬───────┼────────┬──────────────┐
       ▼              ▼       ▼        ▼              ▼
  User Profile   Learning AI  Skill Graph  Content Engine  Analytics
       │              │       │        │              │
       └──────────────┴───────┼────────┴──────────────┘
                              ▼
                  Recommendation Engine
                              │
                              ▼
                  Personalized Learning Feed
```

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4. System Layers

Bloom is composed of six logical layers.

Layer 1 — Presentation Layer

The Presentation Layer is responsible for all learner interactions.

Responsibilities:

- User onboarding
- Dashboard
- Learning feed
- Roadmaps
- Progress visualization
- Guard Mode controls
- AI assistant interface

Possible interfaces include:

- Browser extension
- Web application
- Mobile application (future)

This layer contains no recommendation logic.

Layer 2 — Application Layer

This layer coordinates all product functionality.

Responsibilities:

- Authentication
- User sessions
- Request routing
- Learning workflows
- Progress updates
- API orchestration

It acts as the communication bridge between the user interface and backend intelligence.

Layer 3 — Learning Intelligence Layer

This is Bloom's core intelligence.

It transforms user goals into structured learning experiences.

Major components include:

- Learning Intent Engine
- Skill Graph Engine
- Creator Intelligence
- Content Intelligence
- Learning Experience Orchestrator
- Recommendation Engine
- Guard Engine

Most of Bloom's proprietary logic exists here.

Layer 4 — Knowledge Layer

Responsible for organizing knowledge.

Contains:

- Skill Graph
- Career Graph
- Content Graph
- Creator Graph
- User Learning Graph

These graphs collectively model the learning ecosystem.

Layer 5 — Data Layer

Stores structured and unstructured information.

Includes:

- User profiles
- Learning history
- Content metadata
- Progress
- Analytics
- Recommendation history
- Feedback events

Layer 6 — External Platform Layer

Bloom integrates existing educational ecosystems.

Examples include:

- Video platforms
- Code repositories
- Technical documentation
- Learning communities
- Blogs
- Podcasts
- Research repositories

Bloom consumes these resources.

It does not replace them.

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5. Core Components

User Profile Service

Responsible for maintaining learner information.

Stores:

- Learning Intent Profile
- Career goals
- Preferences
- Progress
- Settings

Every recommendation begins here.

Learning Intent Engine

Transforms learner objectives into machine-readable learning goals.

Responsibilities:

- Interpret onboarding conversations
- Estimate learner level
- Identify missing prerequisites
- Continuously refine learner profile

Outputs a continuously evolving Learning Intent Profile.

Skill Graph Engine

The Skill Graph represents knowledge relationships.

Responsibilities:

- Map prerequisite skills
- Discover future learning paths
- Support multiple career routes

- Track learner progression

Unlike static roadmaps, the graph allows multiple valid learning paths.

Creator Intelligence Engine

Not all educators teach in the same way.

This engine models creators based on characteristics such as:

- Teaching style
- Difficulty level
- Topic expertise
- Depth of explanation
- Audience suitability
- Educational quality

Recommendations consider both content quality and creator suitability.

Content Intelligence Engine

Responsible for understanding educational resources.

Tasks include:

- Metadata extraction
- Topic identification
- Skill tagging
- Quality assessment
- Freshness evaluation
- Duplicate detection

Popularity is only one quality signal.

Recommendation Engine

The central decision-making component.

Inputs:

- Learning Intent Profile
- Skill Graph
- Creator Intelligence

- Content Intelligence
- Learning History
- Feedback Signals

Outputs:

- Personalized learning feed
- Daily learning sessions
- Suggested projects
- Review recommendations

Recommendations should always be explainable.

Learning Experience Orchestrator

Rather than recommending isolated resources, Bloom recommends complete learning experiences.

Example:

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Concept

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Video

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Documentation

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Hands-on Exercise

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Mini Project

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Reflection

↓

Revision
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↓

Next Skill
```

The orchestrator sequences these activities according to the learner's context.

Guard Engine

Responsible for maintaining focus during learning sessions.

Possible actions include:

- Hide distractions
- Replace unrelated recommendations
- Reduce interface noise
- Encourage focused sessions

The learner always retains control.

Analytics Engine

Collects anonymized learning signals.

Tracks:

- Resource completion
- Learning consistency
- Recommendation acceptance
- Progression speed
- Session quality

These signals continuously improve future recommendations.

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6. Data Flow

The system follows a continuous feedback cycle.

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User

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```

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Onboarding

↓

Learning Intent Profile

↓

Skill Graph

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Recommendation Engine

↓

Learning Session

↓

Feedback Signals

↓

Profile Update

↓

Improved Recommendations
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Every interaction contributes to personalization.

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7. Recommendation Pipeline

The recommendation engine follows a structured process.

Step 1

Understand learner context.

Step 2

Locate learner position within the Skill Graph.

Step 3

Determine candidate skills.

Step 4

Retrieve educational resources.

Step 5

Evaluate resource quality.

Step 6

Evaluate creator suitability.

Step 7

Construct learning experience.

Step 8

Deliver personalized recommendations.

Step 9

Observe learner feedback.

Step 10

Refine learner model.

8. Feedback Architecture

Bloom continuously learns from learner behavior.

Signals include:

Positive:

- Resource completed
- Resource bookmarked
- Project completed
- Session completed

Negative:

- Early abandonment
- Resource skipped
- Difficulty mismatch
- Session interruption

These signals improve future recommendations.

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9. Explainability

Every recommendation should answer:

- Why was this recommended?
- Which career goal does it support?
- Which prerequisite does it satisfy?
- What does it unlock next?

Transparency builds learner trust.

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10. Scalability

Bloom should support:

- New career paths
- New educational platforms
- New recommendation models
- Additional AI providers
- Enterprise deployments

New components should integrate without redesigning the system.

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11. Security and Privacy

Core principles:

- Minimal data collection
- User-owned learning history
- Transparent data usage
- Secure authentication
- Encrypted communication
- Permission-based integrations

Bloom should only collect information necessary to improve learning.

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12. Failure Handling

The architecture should degrade gracefully.

Examples:

- If AI services are unavailable, serve cached recommendations.
- If external content is unavailable, recommend alternative resources.
- If personalization is limited, fall back to curated learning paths.
- If recommendation confidence is low, explain the uncertainty rather than making arbitrary suggestions.

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13. Architectural Decisions

Decision	Rationale
Modular architecture	Independent evolution of components
Graph-based knowledge model	Represents complex learning dependencies
AI-assisted personalization	Adapts recommendations to individuals
Platform-independent design	Supports multiple learning ecosystems
Explainable recommendations	Builds learner trust
Continuous feedback loop	Improves recommendation quality over time

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14. Long-Term Evolution

The architecture is designed to evolve through four stages.

Stage 1

Single-platform learning assistant.

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Stage 2

Multi-platform learning intelligence.

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Stage 3

Personalized lifelong learning system.

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Stage 4

Learning Intelligence Platform.

At the final stage, Bloom becomes infrastructure that powers personalized learning experiences for individuals, educational institutions, and enterprise learning systems.

Architecture Summary

Bloom is built as a layered, modular, AI-powered architecture that separates user interaction, learning intelligence, knowledge representation, recommendation generation, and data management. This separation enables the platform to scale across educational domains, integrate new learning sources, and continuously improve personalization while maintaining explainability, extensibility, and learner trust.

The architecture is intentionally designed so that **the intelligence—not the content—is the product**.